MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours) REGULAR

EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Core Course (DSC) - MG5DSCFSC301 - FORENSIC TOXICOLOGY

(2024 ADMISSION ONWARDS)

SCHEME OF VALUATION

Duration: 1 hour 30 minutes **Maximum Marks: 50**

General instructions: Award marks for equivalent scientifically correct answers. The points below are indicative, not exhaustive. For questions with multiple parts, award marks proportionately. Do not exceed the maximum mark for each question.

Part A – Short Answer Type Questions

Answer any 10 questions. Each question carries 2 marks. Maximum: 20 marks.

1. Mention two examples of antidotes used in poisoning cases. (2 marks)

- Any two appropriate examples, e.g., naloxone for opioid toxicity; atropine plus pralidoxime for organophosphate poisoning — 1 mark
- One further correct antidote/toxin pairing, e.g., N-acetylcysteine for paracetamol; deferoxamine for iron — 1 mark

2. A forensic sample is suspected to contain a phosphide compound. Which preliminary chemical test would you select and what is its purpose? (2 marks)

- A suitable preliminary test for phosphide/phosphine, such as a validated silver-nitrate paper test for liberated phosphine, may be selected — 1 mark
- Purpose is presumptive screening; specify safe, controlled handling and confirmatory analysis as appropriate — 1 mark

3. What is meant by evidence material in food-poisoning investigation? (2 marks)

- Materials collected to investigate suspected food poisoning, e.g., leftover food, vomitus, gastric contents, stool, blood or suspected container — 1 mark
- Proper labelling, preservation, representative sampling and chain of custody are essential — 1 mark

4. Mention two signs of carbon monoxide poisoning. (2 marks)

- Any two: headache, dizziness, nausea, weakness, confusion, syncope — 1 mark
- Severe cases may show seizures/coma; cherry-red colour is uncommon and should not be relied on — 1 mark

5. How can the Dithionite test be applied to Paraquat or Diquat examination? (2 marks)

- Sodium dithionite in alkaline conditions reduces paraquat/diquat to coloured radical species (paraquat commonly blue) — 1 mark
- Presumptive screening only; confirm with a specific validated instrumental method — 1 mark

6. What are plant poisons? (2 marks)

- Toxic substances naturally occurring in plants — 1 mark
- Examples include alkaloids, glycosides or toxic plant constituents such as oleandrin/odollam-related cardiac glycosides — 1 mark

7. Differentiate between the analysis of ethyl alcohol in beverages and biological fluids. (2 marks)

- Beverages: usually direct measurement of ethanol in the drink, with matrix/dilution considerations — 1 mark
- Biological fluids: validated forensic method (often headspace GC) with matrix effects, preservation, calibration and interpretation considerations — 1 mark

8. Explain the use of spectrophotometry in poison analysis. (2 marks)

- Measures absorption of light at selected wavelengths to estimate analyte concentration or support identification — 1 mark
- Useful for suitable compounds/reactions; selectivity is limited and confirmation may be needed — 1 mark

9. Why are instrumental techniques used for identification? (2 marks)

- Provide greater sensitivity, selectivity and reproducibility than many simple screening tests — 1 mark
- Can separate and identify compounds in mixtures and/or quantify them using validated methods — 1 mark

10. Analyse how the route of ingestion can influence the absorption and distribution of a poison. (2 marks)

- Route affects absorption rate/bioavailability; oral exposure may be affected by dissolution, food and first-pass metabolism — 1 mark
- Distribution/onset also depend on dose, chemical properties, tissue perfusion and route-specific exposure — 1 mark

11. How can HPLC be applied to poison detection? (2 marks)

- Separates components in a sample based on interactions with stationary and mobile phases — 1 mark
- Detector response/retention time, preferably with suitable confirmatory detection and standards, supports identification/quantification — 1 mark

12. Explain one limitation of a toxicological method. (2 marks)

- Any valid limitation: matrix interference, limited sensitivity, lack of specificity, detection-window limits, degradation or inability to distinguish exposure from impairment — 1 mark
- Explain its effect or need for controls/confirmation/validated interpretation — 1 mark

Part B – Short Essay Type Questions

Answer any 5 questions. Each question carries 4 marks. Maximum: 20 marks.

13. Explain the principle and forensic application of the Conway diffusion method for the determination of ethyl alcohol. (4 marks)

- Volatile ethanol diffuses from the outer chamber of a sealed Conway microdiffusion unit — 1 mark
- Ethanol is absorbed in a suitable receiving reagent in the inner chamber — 1 mark
- The captured ethanol is determined by a validated chemical/analytical procedure with standards and blanks — 1 mark
- Used for determination of ethanol in appropriate samples; limitations include diffusion conditions, interference and need for validation — 1 mark

14. Analyse the role of preliminary colour tests in forensic toxicology and explain why their results require confirmation by more specific analytical techniques. (4 marks)

- Rapid, inexpensive screening of suspected drugs/poisons — 1 mark
- Colour reaction can indicate a class or possible analyte, not definitive identity — 1 mark
- Cross-reactions, matrix effects and subjective colour interpretation can produce false positives/negatives — 1 mark
- Confirm using validated specific methods such as GC–MS or LC–MS/MS, as appropriate — 1 mark

15. Describe insect and animal toxins relevant to toxicological analysis. (4 marks)

- Examples of insect toxins/venoms: bee, wasp, ant or other medically relevant venom exposures — 1 mark
- Examples of animal toxins: snake venom, scorpion venom, marine toxins or toxic secretions — 1 mark
- Discuss relevant toxic components/effects, e.g., neurotoxic, haemotoxic or cytotoxic action — 1 mark
- Specimen selection, clinical/history context and validated targeted analysis are important — 1 mark

16. Analyse how clean-up and concentration procedures improve the reliability of the subsequent identification of drugs and poisons. (4 marks)

- Remove proteins, lipids and other matrix interferences — 1 mark
- Concentrate trace analytes to improve detection limits — 1 mark
- Improve chromatographic separation, detector response and reduce instrument contamination — 1 mark
- Use recovery checks/internal standards and avoid analyte loss or contamination during preparation — 1 mark

17. Explain TLC examination of barbiturates and cannabinoids. (4 marks)

- Extract the sample and apply extract plus reference standards to a suitable TLC plate — 1 mark
- Develop in a validated solvent system and allow separation — 1 mark
- Visualize using appropriate detection reagents/UV and compare Rf values and appearance with standards — 1 mark
- TLC is presumptive; matrix effects and similar Rf values mean confirmation by a specific method is required — 1 mark

18. Discuss the presentation of laboratory findings in a toxicological report. (4 marks)

- Identify case/sample details, seals, condition, receipt and chain of custody — 1 mark
- State methods, controls, calibration/quality assurance and relevant limitations — 1 mark
- Report results with units, detection/quantification limits where relevant and clear distinction between detected/not detected — 1 mark
- Give cautious interpretation, avoid unsupported conclusions, and include analyst/date/signature as required — 1 mark

19. Analyse the factors that affect the extraction of acidic and basic drugs from viscera, blood, and urine. (4 marks)

- pH and analyte pKa determine ionisation and partitioning — 1 mark
- Solvent polarity, liquid–liquid or solid-phase extraction conditions and solvent-to-sample ratio — 1 mark
- Sample matrix, protein binding, metabolites, degradation and storage/preservation — 1 mark
- Extraction recovery, selectivity, internal standards and validation affect reliability — 1 mark

Part C – Essay Type Questions

Answer any 1 question. Each question carries 10 marks. Maximum: 10 marks.

20. Explain the general principles and procedures involved in the extraction and isolation of poisons from biological specimens. Discuss the factors affecting the recovery of poisons. (10 marks)

- Selection and documentation of specimens based on case history and suspected poison; preserve chain of custody — 2 marks
- Preliminary examination/screening and appropriate sample homogenisation or aliquoting — 2 marks
- Suitable extraction/clean-up strategy (e.g., liquid–liquid, solid-phase, headspace or other method chosen for analyte properties) — 2 marks
- Isolation/concentration, use of blanks, standards/internal standards and recovery controls — 2 marks
- Factors: pH/pKa, polarity, binding, matrix, degradation, storage, extraction solvent, technique and analyte volatility; confirm and quantify with validated methods — 2 marks

21. Analyse the differences between chemical and bacterial food poisoning with respect to their toxicological examination. (10 marks)

- Chemical food poisoning arises from chemicals/toxins such as pesticides, metals or other toxic substances; bacterial food poisoning arises from pathogenic bacteria or their toxins — 2 marks
- Case history, symptom onset, food history and epidemiological pattern guide investigation — 2 marks
- Chemical examination: targeted screening/confirmation and quantification in food and biological specimens using validated chemical/instrumental methods — 2 marks
- Bacterial examination: microbiological culture, molecular methods and/or toxin assays as appropriate; collect suitable food and clinical specimens — 2 marks
- Compare evidence, controls, sampling and interpretation; consider preformed toxins, incubation period, specimen degradation and avoid overclaiming from a single positive result — 2 marks

Marking note: The paper instructs students to attempt at least one question from each course outcome. Apply the examination regulations when evaluating attempt requirements.